Path-Specific Beamforming and Nonuniform Doppler Compensation for Underwater Acoustic Communications

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***Abstract*—This article introduces methods for path-specific beamforming and adaptive compensation of nonuniform Doppler drifts in single-carrier broadband underwater acoustic communications. We focus on scenarios where different propagation paths experience varying Doppler drifts, which can significantly degrade conventional receivers. The proposed method employs broadband beamforming to separate the multipath components. Each path is then individually time-aligned and Doppler-synchronized before being processed by a multichannel fractionally spaced decision-feedback equalizer (DFE). By time-aligning the signals before equalization, the DFE filters need only cover the dispersion of each isolated path rather than the entire multipath spread, which leads to an overall reduction in the equalizer size. To account for time compression and dilation, we incorporate a delay-locked loop alongside a phase-locked loop to perform adaptive online resampling or delay tracking, thus eliminating the need for front-end resampling even in high-Doppler scenarios. Simulation and experimental results demonstrate that the proposed approach operates successfully under conditions that cause conventional receivers to fail. In cases where the conventional receiver performs well, our proposed algorithm achieves comparable performance with significantly fewer training symbols, yielding excellent performance in terms of data detection mean squared error and bit error rate.**

***Index Terms*—Adaptive resampling, broadband acoustic beamforming, decision-feedback equalizer (DFE), path-specific Doppler, underwater acoustic communications.**

I. Introduction

A DISTINGUISHING characteristic of underwater acoustic communication channels is that signals propagating over multiple paths can experience different fading, Doppler distortions, and delays. Path-specific Doppler distortions or drifts can be encountered in practical situations where the relative motion between the transmitter and receiver causes some paths to lengthen while others to shorten [1], [2], [3], [4]. This motion results in signal dilation or compression. Specifically, a transmitted signal of duration T is observed at the receiver as a superposition of multiple path signals, each having duration $T(1 + a_p)$, where $a_p = v_{tr,p}/c$ is the Doppler scaling factor of the pth path, which is related to the corresponding relative transmitter–receiver velocity $v_{tr,p}$ and speed of propagation c.

The first step in mitigating the Doppler effect is typically front-end resampling, where a common Doppler scaling factor is estimated from synchronization preambles and applied to resample the received signal. This initial synchronization compensates for a dominant Doppler factor but inevitably leaves a residual distortion due to estimation error and path-dependent Doppler. After downconversion, the residual Doppler remains evident as phase and delay drifting. In single-carrier systems, the phase drift is typically tracked using a phase-locked loop (PLL) in combination with a decision-feedback equalizer (DFE) [5], [6], [7], [8], [9], while residual delay drift is often ignored, as its impact is minimal over typical data packet lengths of a few thousand bits.

Although neglecting the delay drift is justified in many scenarios, significant path-specific Doppler variations or extreme Doppler conditions require special consideration. Such situations occur either at very high speeds or in communication at regular speeds but with very long data packets.

Prior work on mitigating Doppler effects in underwater acoustic communications includes several approaches. Sharif et al. [10] used correlation techniques to estimate the Doppler factor and employed sample rate conversion for resampling. In [11], a recursive algorithm was developed to track the Doppler factor over time. The authors in [12] and [13] proposed simplified versions of the cross-ambiguity function method for coarse Doppler estimation. These approaches generally assume that Doppler effects remain constant throughout the duration of a data packet, limiting their effectiveness to short packets with frequent synchronization preambles.

Choi et al. [14] proposed an online method for estimating nonuniform Doppler drifts across multiple propagation paths. In [15], the cyclostationarity properties of communication signals were exploited to estimate path-specific Doppler scaling factors. A time-reversal-based Doppler compensation algorithm for single-carrier systems was also developed in [16]. However, these methods remain sensitive under highly time-varying conditions and tend to degrade in performance over long transmission durations.

Adaptive multichannel equalization has traditionally been used for high-rate communication over Doppler-distorted underwater acoustic channels [5], but its computational complexity

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increases significantly with the number of receiving elements. In a subsequent work, Stojanovic et al. [17] implemented a beamformer to separate the multiple paths, which not only led to a lower complexity receiver but also achieved a performance similar to the one delivered by the conventional multichannel receiver. Following this idea, Blair and Preisig [18] explored beamforming approaches that relied on angles of arrival computed from the experimental channel geometry. In addition, beamforming with spatial clustering in the context of orthogonal frequency division multiplexing, as well as beamforming for very-high-frequency acoustic channels, has been addressed in [19], [20], and [21], respectively. However, none of these studies focus on tracking path-specific Doppler drifts.

Building on beamforming, we address the problem of path-specific Doppler tracking [22]. Specifically, we propose a system that includes a single transmitter and a receiver equipped with a uniform linear array, where beamforming is used to isolate multiple propagation paths. The signals arriving over different paths are time-aligned and fed to a DFE, which tracks the Doppler-induced phase for each path individually. Compared to the conventional multichannel equalizer, which requires long adaptive filters, inclusion of front-end beamforming enables the use of shorter equalizer filters.

In addition, we address scenarios involving extreme path-specific Doppler over extended periods. We employ not only an individual PLL for each isolated path, but also an additional delay-locked loop (DLL) as well. The DLL is coupled with the PLL and capable of performing adaptive online resampling. The adaptively resampled paths are then combined using a multichannel DFE to detect the data symbols. The inclusion of a DLL eliminates the need for front-end resampling, even in high-Doppler regimes [23]. Such regimes involve communication at high speeds (e.g., on the order of 10 m/s) or communication at regular speeds but with long data packets. The remaining structure of the DFE is similar to that described in [5] and consists of a bank of adaptive feedforward filters (one per propagation path), followed by a decision-feedback filter.

In summary, the proposed algorithm leverages plane-wave propagation under conditions of full path coherence across an array of elements and is designed to operate in path-specific Doppler regimes. The algorithm consists of two main stages: 1) beamforming to separate propagation paths and 2) adaptive online resampling to track Doppler drifts driven by a per-path PLL. When path coherence is insufficient, front-end beamforming cannot be applied; in this case, the algorithm employs adaptive online resampling together with a conventional multichannel DFE driven by a single PLL. The design concepts are demonstrated through simulation of a shallow-water channel, as well as in experimental underwater transmissions, yielding excellent system performance in terms of data detection mean squared error (MSE) and bit error rate (BER).

The rest of this article is organized as follows. We introduce the signal and channel model in Section II, and present the beamforming and Doppler tracking algorithm in Section III. The adaptive online resampling method is introduced in Section IV, and the numerical results are presented in Section V. In Section VI, we present the experimental results obtained from the Surface Processes Acoustic Communications Experiment (SPACE), the Mobile Acoustic Communications Experiment (MACE), and the Kauai AComms Multidisciplinary University (MURI) (KAM) experiment. Among the three experiments considered, the SPACE and KAM data sets exhibit sufficient path coherence for beamforming-based path separation, enabling application of the path-based receiver architecture. In contrast, the MACE data set lacks path coherence across the array, and thus, the receiver implements the multichannel conventional DFE aided by adaptive online resampling. Finally, Section VII concludes this article.

II. Signal and Channel Model

We consider a single-input multiple-output single-carrier system with an array of M equally spaced receiving elements. The transmitted passband signal is given by

$$s(t) = \text{Re}\left\{\sum_k d[k]g(t-kT)e^{j2\pi f_c t}\right\} \tag{1}$$

where $d[k]$ denotes the data symbol transmitted during the kth signaling interval of duration $T = 1/R$, where R is the symbol rate, $g(t)$ represents the response of the transmitter shaping filter, e.g., a raised cosine filter with roll-off factor of α_{rc}, and f_c the center frequency. The signal observed on the mth receiving element is given by

$$r_m(t) = \bar{r}_m(t) + n_m(t), \quad m = 0, \ldots, M-1 \tag{2}$$

where

$$\bar{r}_m(t) = \int_{-\infty}^{\infty} h_m(\tau, t)s(t-\tau)d\tau \tag{3}$$

and $h_m(\tau, t)$ and $n_m(t)$ represent the time-varying channel impulse response and the additive noise seen on the mth element, respectively. The channel response at time t is represented as a function of delay τ as

$$h_m(\tau, t) = \sum_{p=0}^{P-1} h_p^m(t)\delta\left(\tau - \tau_p^m(t)\right) \tag{4}$$

where $h_p^m(t), \tau_p^m(t)$, and P represent the path gains, path delays, and number of propagation paths, respectively. We assume that the path gains vary slowly across the duration of the signal and are approximately the same across the array, i.e., $h_p^m(t) \approx h_p$, and that path delays can be further modeled as

$$\tau_p^m(t) = \tau_p^m + \epsilon_p(t), \quad m = 0, \ldots, M-1 \tag{5}$$

where $\epsilon_p(t)$ represents the time-varying component of the delay or delay drift. After time synchronization and downshifting, the received baseband signal can be modeled as

$$\tilde{v}_m(t) = \bar{v}_m(t) + \tilde{w}_m(t), \quad m = 0, \ldots, M-1 \tag{6}$$

where

$$\bar{v}_m(t) = \sum_k d[k] \sum_{p=0}^{P-1} h_p e^{-j2\pi f_c \tau_p^m} \times g(t - kT - \tau_p^m - \epsilon_p(t))e^{-j2\pi f_c \epsilon_p(t)} \tag{7}$$

and $\tilde{w}_m(t)$ is the additive complex-baseband noise. The signals (6) are used as inputs to the front-end beamformer.

III. Beamforming and Path-Specific Doppler Tracking

Assuming a plane wave propagation model, the constant component of the delays (5) can be expanded as

$$\tau_p^m = \tau_p^0 + m\Delta\tau_p \tag{8}$$

where τ_p^0 is the delay observed on the first array element, and $\Delta\tau_p = \frac{\delta}{c}\sin\theta_p$ is the incremental delay across the array given in terms of the interelement spacing δ, the sound speed c (nominally 1500 m/s), and the angle of arrival θ_p corresponding to path p.

During a short time interval (e.g., the duration of a preamble signal), $\epsilon_p(t)$ may introduce a small delay perturbation, i.e., $t \approx t + \epsilon_p(t)$. Assuming a sampling rate of $f_s = 1/T_s = N_s/T$, where $N_s \geq 2$ is an integer number of samples per symbol, the signal samples are expressed as

$$\tilde{v}_m[n] = \sum_k d[k] \sum_{p=0}^{P-1} c_p e^{-j2\pi m f_c \Delta\tau_p} \times g\left[n - k\frac{T}{T_s} - \frac{\tau_p^0}{T_s} - m\frac{\Delta\tau_p}{T_s}\right] + \tilde{w}_m[n] \tag{9}$$

for $m = 0, \ldots, M-1$, where $c_p = h_p e^{-j2\pi f_c \tau_p^0}$ represents the baseband equivalent path gain. Applying an L-point normalized discrete Fourier transform (DFT) to (9), we obtain the equivalent frequency-domain representation

$$\tilde{V}_m[\ell] = \sum_k d[k] G[\ell] e^{-j2\pi\frac{\ell k T}{T_s}} \sum_{p=0}^{P-1} c_p e^{-j2\pi m f_\ell \Delta\tau_p} + \tilde{W}_m[\ell], \quad \ell = 0, \ldots, L-1 \tag{10}$$

where $G[\ell]$ and $\tilde{W}_m[\ell]$ are the frequency-domain representations of the shaping filter and the noise, respectively, and

$$f_\ell = \begin{cases} f_c + \ell\Delta f, & 0 \leq \ell \leq \frac{L}{2} \\ f_c + (\ell - L)\Delta f, & \frac{L}{2} + 1 \leq \ell \leq L-1 \end{cases} \tag{11}$$

represents the ℓth frequency bin, where $\Delta f = (LT_s)^{-1}$ is the frequency bin spacing. The frequency-domain coefficients (10) are stacked into the vector

$$\tilde{\mathbf{V}}[\ell] = \sum_k d[k] G[\ell] e^{-j2\pi\frac{\ell k T}{T_s}} \sum_{p=0}^{P-1} c_p \mathbf{s}_M(2\pi f_\ell \Delta\tau_p) + \tilde{\mathbf{W}}[\ell], \quad \ell = 0, \ldots, L-1 \tag{12}$$

where

$$\tilde{\mathbf{V}}[\ell] \triangleq [\tilde{V}_0[\ell] \;\; \tilde{V}_1[\ell] \;\; \ldots \;\; \tilde{V}_{M-1}[\ell]]^\top \in \mathbb{C}^{M\times 1}$$
$$\tilde{\mathbf{W}}[\ell] \triangleq [\tilde{W}_0[\ell] \;\; \tilde{W}_1[\ell] \;\; \ldots \;\; \tilde{W}_{M-1}[\ell]]^\top \in \mathbb{C}^{M\times 1}$$
$$\mathbf{s}_M(\chi) \triangleq [1 \;\; e^{-j\chi} \;\; \ldots \;\; e^{-j(M-1)\chi}]^\top \in \mathbb{C}^{M\times 1}. \tag{13}$$

The vector $\mathbf{s}_M(\chi)$ is referred to as the steering vector.

A. Null Steering

We can express (12) in a compact form as

$$\tilde{\mathbf{V}}[\ell] = \mathbf{S}_\ell \mathbf{V}[\ell] + \tilde{\mathbf{W}}[\ell], \quad \ell = 0, 1, \ldots, L-1 \tag{14}$$

where

$$\mathbf{S}_\ell \triangleq [\mathbf{s}_M(2\pi f_\ell \Delta\tau_0) \;\; \mathbf{s}_M(2\pi f_\ell \Delta\tau_1) \;\; \ldots \;\; \mathbf{s}_M(2\pi f_\ell \Delta\tau_{P-1})] \in \mathbb{C}^{M\times P}$$
$$\mathbf{V}[\ell] \triangleq U[\ell]\mathbf{c}_\ell \in \mathbb{C}^{P\times 1}$$
$$\mathbf{c}_\ell \triangleq [c_0 e^{-j2\pi\ell\Delta f \tau_0^0} \;\; c_1 e^{-j2\pi\ell\Delta f \tau_1^0} \;\; \ldots \;\; c_{P-1} e^{-j2\pi\ell\Delta f \tau_{P-1}^0}]^\top \in \mathbb{C}^{P\times 1}$$
$$U[\ell] \triangleq \sum_k d[k] G[\ell] e^{-j2\pi\frac{\ell k T}{T_s}} \tag{15}$$

where $G[\ell]$ is the normalized DFT of $g(nT_s)$. A beamforming matrix is applied to obtain the estimated path signals[1]

$$\hat{\mathbf{V}}[\ell] = \begin{cases} \mathbf{\Phi}'_\ell \tilde{\mathbf{V}}[\ell], & \ell \in \mathcal{L}_B \\ 0, & \text{otherwise} \end{cases} \tag{16}$$

where

$$\mathcal{L}_B = \left\{ \ell \mid 0 \leq \ell \leq \bar{L} \;\; \text{and} \;\; L - \bar{L} \leq \ell \leq L-1 \right\}$$

is the set of frequency indices within the bandwidth of the signals in (14), and

$$\bar{L} = \left\lceil \frac{L(1+\alpha_{rc})}{2N_s} \right\rceil$$

is the number of frequency bins required to cover one half of the occupied band.

The zero-forcing or null-steering matrix is given by

$$\mathbf{\Phi}_\ell = \mathbf{S}_\ell \left[\mathbf{S}'_\ell \mathbf{S}_\ell\right]^{-1} \tag{17}$$

which is contingent upon having knowledge of the true angles of arrival θ_p. In practical situations, these angles need to be estimated.

B. Angle Estimation

Estimation of the angles of arrival is performed by applying a steering operation across (14) and forming the metric

$$S(\theta) = \sum_{\ell \in \mathcal{L}_B} \left| \mathbf{s}'_M(2\pi f_\ell \Delta\tau) \tilde{\mathbf{V}}[\ell] \right|^2 \tag{18}$$

where $\Delta\tau = (\delta/c)\sin\theta$ is the hypothesized incremental delay. The path angles θ_p are now estimated as the local maxima

$$\hat{\theta}_p = \text{peaks}_\theta \left\{ S(\theta) \right\} \tag{19}$$

where the term $\text{peaks}_\theta\{\cdot\}$ denotes the operation that identifies the values of θ for which the local maxima of the function $S(\theta)$ are attained.

The estimated null-steering matrix $\hat{\mathbf{\Phi}}_\ell$ is constructed by replacing the unknown incremental delays $\Delta\tau_p$ by their estimates

[1] $(\cdot)'$ denotes conjugate transpose.

$\hat{\Delta}\tau_p = (\delta/c)\sin\hat{\theta}_p$, which are obtained from the angle estimates $\hat{\theta}_p$. The estimated null-steering matrix can be expressed as

$$\hat{\mathbf{\Phi}}_\ell = \hat{\mathbf{S}}_\ell \left[\hat{\mathbf{S}}'_\ell \hat{\mathbf{S}}_\ell\right]^{-1}, \quad \ell \in \mathcal{L}_B \tag{20}$$

where

$$\hat{\mathbf{S}}_\ell \triangleq \left[\mathbf{s}_M(2\pi f_\ell \hat{\Delta}\tau_0) \quad \ldots \quad \mathbf{s}_M(2\pi f_\ell \hat{\Delta}\tau_{P-1})\right] \in \mathbb{C}^{M\times P}. \tag{21}$$

The pth column of $\hat{\mathbf{\Phi}}_\ell$ corresponds to a beamforming vector that steers a beam toward a path angle $\hat{\theta}_p$ while simultaneously places nulls in the directions of other paths.

There may be situations where the path angles $\hat{\theta}_p$ are close to each other, causing the matrix $\mathbf{S}'_\ell\mathbf{S}_\ell$ to become ill-conditioned, i.e., some of its eigenvalues to become very small or approximately zero. In this case, applying null-steering would lead to significant noise amplification after beamforming which would degrade overall system performance.

A common remedy found in the literature for ill-conditioning is diagonal loading (or regularization) [24]. The diagonally loaded null-steering matrix is given by

$$\hat{\mathbf{\Phi}}_{\ell,\mathrm{DL}} = \hat{\mathbf{S}}_\ell \left[\hat{\mathbf{S}}'_\ell \hat{\mathbf{S}}_\ell + \gamma \mathbf{I}_P\right]^{-1}, \quad \ell \in \mathcal{L}_B \tag{22}$$

where $\gamma > 0$ and $\mathbf{I}_P$ is a $P \times P$ identity matrix. Although diagonal loading alleviates numerical instabilities, it introduces a tradeoff between imperfect interpath interference suppression and reduced noise amplification.

Another approach to mitigate ill-conditioning is to perform beamforming using only high-energy well-separated paths. However, this approach may again come at the cost of reduced interpath suppression performance. In Section V, we present results for both approaches and demonstrate that either can provide satisfactory performance.

C. Time-Alignment of Paths

After beamforming, the contribution of the pth multipath signal seen on frequency f_ℓ is given by

$$\hat{V}_p[\ell] = \sum_k d[k] c_p e^{-j2\pi\ell\Delta f \tau_p^0} G[\ell] e^{-j2\pi\frac{k\ell T}{T_s}} + \hat{W}_p[\ell], \quad p = 0,\ldots,P-1, \quad \ell \in \mathcal{L}_B. \tag{23}$$

These signals are the normalized DFT coefficients of the pth multipath signal $\hat{v}_p[n]$, which contains the samples of the pth beamformed signal $\hat{v}_p(t)$. If we consider now the delay drift $\epsilon_p(t)$, then the beamformed signal is approximately given by

$$\hat{v}_p(t) \approx \sum_k d[k] c_p g(t - \tau_p^0 - \epsilon_p(t) - kT) e^{-j2\pi f_c \epsilon_p(t)} + \hat{w}_p(t), \quad p = 0,\ldots,P-1 \tag{24}$$

where $\hat{w}_p(t)$ is the residual noise on the pth beamformed path.

In a conventional receiver system, where M signals (6) are fed directly to a multichannel DFE, the equalizer field of view needs to cover the multipath spread of the channel. In contrast, in a system that implements front-end beamforming, each of the P signals (24) can be individually delayed by τ_p^0, i.e., $v_p(t) = \hat{v}_p(t + \tau_p^0)$, before being fed to the equalizer. Consequently, the equalizer only needs to compensate for a small dispersion on each isolated path. The process of extracting the time-aligned path signals $v_p(t)$ is illustrated in Fig. 1.

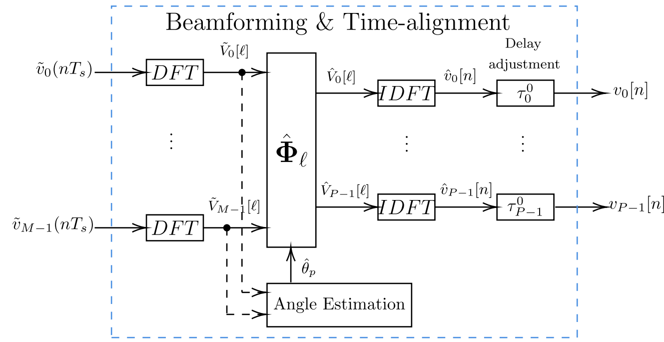


Fig. 1. Front-end beamforming diagram.

IV. Adaptive Online Resampling

In this section, we introduce an adaptive resampling operation, which performs adaptive delay tracking using a DLL to compensate for the path-specific Doppler drift.

A. Adaptive Delay Tracking

The time-aligned signal arriving on path p can be represented as

$$v_p(t) = \sum_k d[k] g_p(t - kT - \epsilon_p(t)) e^{-j2\pi f_c \epsilon_p(t)} + w_p(t) \tag{25}$$

where $g_p(t) = c_p g(t)$ is the time-aligned and drift-free response of the pth path, and $w_p(t) = \hat{w}(t + \tau_p^0)$ is the noise. The delay drift can be expressed as

$$\epsilon_p(t) = a_p(t) \cdot t \tag{26}$$

where

$$a_p(t) = \frac{v_{tr,p}(t)}{c} \tag{27}$$

is the Doppler scaling factor, and $v_{tr,p}(t)$ is the relative transmitter/receiver velocity projected onto the pth path. In practice, the Doppler scaling factor changes slowly compared to the sample interval T_s; therefore, $a_p(t)$ can be approximated as a piecewise constant $a_p(t) \approx a_{p,n}$ for $t \in \mathcal{T}_n$, where $\mathcal{T}_n$ is an interval of time around t_n, e.g., $\mathcal{T}_n \in [t_{n-1}, t_n)$, and $a_{p,n} \approx a_{p,n-1}$. To compensate for the time dilation/compression in the interval $\mathcal{T}_n$, a time-scaling operation must be applied to the signal $v_p(t)$ to obtain

$$v_p\left(\frac{t_n}{1 - a_{p,n}}\right) = \sum_k d[k] g_p(t_n - kT) e^{j\varphi_p(t_n)} + \xi_p(t_n) \tag{28}$$

where

$$\varphi_p(t_n) = -2\pi f_c \frac{a_{p,n}}{1 - a_{p,n}} t_n \tag{29}$$

and $\xi_p(t_n)$ represents the noise contribution. Defining the interval of time $t_n = nT$, the time-scaled signal (28) can also be written as

$$v_p\left(nT - \frac{\varphi_p(nT)}{2\pi f_c}\right) = \sum_k d[k] g_p(nT - kT) e^{j\varphi_p(nT)} + \xi_p(nT) \tag{30}$$

where the basic idea of adaptive delay tracking rests on the notion that if the time-scaled signals (30) were available, they could be fed to the PLL-aided equalizer, whose side product would include the estimated data symbols $\hat{d}[n]$ and the estimated phases $\hat{\varphi}_p(nT)$. These phases could then be used to generate the time-scaled signals for the next symbol interval. The process would then repeat, thus closing the loop between time scaling (delay tracking or adaptive resampling) and equalization/phase tracking.

A possible approach to implementing the time-scaling operation is through linear interpolation between two adjacent signal samples. Using the estimated phase $\hat{\varphi}_p(nT)$ instead of the true phase $\varphi_p(nT)$, the linear resampling operation is given by

$$v_p\left(nT - \frac{\varphi_p(nT)}{2\pi f_c}\right) \approx \mathcal{I}\left\{v_p\left(nT - \frac{\hat{\varphi}_p(nT)}{2\pi f_c}\right)\right\} \tag{31}$$

where $\mathcal{I}\{\cdot\}$ represents the linear interpolation function

$$\mathcal{I}\left\{v_p(t[n])\right\} = (1 - \alpha[n])\, v_p\left(t_L[n]\right) + \alpha[n] v_p\left(t_R[n]\right) \tag{32}$$

where $t_L[n] = \lfloor (t[n])/T_s \rfloor T_s$ and $t_R[n] = \lceil (t[n])/T_s \rceil T_s$ with $\alpha[n] = (t[n] - t_L[n])/T_s$.

B. DFE With Adaptive Delay Tracking

A multichannel fractionally spaced DFE, implemented with spacing $T_s = T/2$, is employed to combine the signals $v_p(t)$, $p = 0, \ldots, P-1$. In this case, two input samples of $v_p(t)$ are resampled in every updating interval T. During the nth symbol interval, these samples can be calculated by using linear interpolation as before

$$y_p\left(nT + i\frac{T}{2}\right) = \mathcal{I}\left\{v_p\left(nT + i\frac{T}{2} - \frac{\hat{\varphi}_p(nT)}{2\pi f_c}\right)\right\} \tag{33}$$

for $i = N_1$ and $N_1 - 1$, where N_1 is the number of causal feedforward taps. Note that there is one phase estimate $\hat{\varphi}_p(nT)$ used to generate both samples. While the two new samples are stacked on the top of an existing vector $\mathbf{y}_p[n-1]$, the two bottom elements of $\mathbf{y}_p[n-1]$ are removed (represented by $\mathbf{y}_p[n-1]_{1:N_f-2}$), and the current vector $\mathbf{y}_p[n]$ of length N_f is formed

$$\mathbf{y}_p[n] = \begin{bmatrix} y_p\left(nT + N_1\frac{T}{2}\right) \\ y_p\left(nT + N_1\frac{T}{2} - \frac{T}{2}\right) \\ \mathbf{y}_p[n-1]_{1:N_f-2} \end{bmatrix}, \quad p = 0, \ldots, P-1 \tag{34}$$

where N_f is the number of feedforward filter coefficients.

The multichannel DFE operations are determined by Stojanovic et al. [5], where a feedforward filter $\mathbf{a}_p[n]$ of length N_f is applied to the signal (34) and phase-shifted by $\hat{\varphi}_p(nT)$. These operations produce an output specified by $x_p[n] = \mathbf{a}_p'[n]\mathbf{y}_p[n]e^{-j\hat{\varphi}_p(nT)}$. The contributions from all P paths are subsequently combined to obtain

$$x[n] = \sum_{p=0}^{P-1} x_p[n] = \sum_{p=0}^{P-1} \mathbf{a}_p'[n]\mathbf{y}_p[n]e^{-j\hat{\varphi}_p(nT)}. \tag{35}$$

The feedback equalizer filter $\mathbf{b}[n]$, which contains N_b feedback coefficients, operates on the vector of symbol decisions $\tilde{\mathbf{d}}[n] = [\tilde{d}[n-1] \quad \tilde{d}[n-2] \quad \ldots \quad \tilde{d}[n-N_b]]^\top$ to produce $z[n] = \mathbf{b}'[n]\tilde{\mathbf{d}}[n]$. The decision $\tilde{d}[n] = \text{decision}\{\hat{d}[n]\}$ is formed by quantizing the estimate $\hat{d}[n]$ to the nearest symbol value. The estimate of the data symbol is then obtained as

$$\hat{d}[n] = x[n] - z[n] \tag{36}$$

and the phases $\hat{\varphi}_p(nT)$ are updated using a digital PLL

$$\hat{\varphi}_p(nT + T) = \hat{\varphi}_p(nT) + K_{f_1}\psi_p[n] + K_{f_2}\sum_{i=0}^{n}\psi_p[i], \quad p = 0, \ldots, P-1 \tag{37}$$

where

$$\psi_p[n] = \text{Im}\left\{x_p[n][\tilde{d}[n] + z[n]]^*\right\} \tag{38}$$

and K_{f_1} and K_{f_2} represent the PLL tracking coefficients. The phase estimates should be initialized as $\hat{\varphi}_p(0) = 0$, $p = 0, \ldots, P-1$.

Equivalently, the symbol estimate $\hat{d}[n] = \mathbf{c}'[n]\mathbf{u}[n]$ can be obtained by forming a composite vector of the feedforward and feedback equalizer filter taps

$$\mathbf{c}[n] = \left[\mathbf{a}_0^\top[n] \quad \ldots \quad \mathbf{a}_{P-1}^\top[n] \quad -\mathbf{b}^\top[n]\right]^\top \tag{39}$$

and the corresponding composite vector of signal samples

$$\mathbf{u}[n] = \left[\mathbf{y}_0[n]e^{-j\hat{\varphi}_0(nT)} \;\; \mathbf{y}_1[n]e^{-j\hat{\varphi}_1(nT)} \;\; \ldots \;\; \mathbf{y}_{P-1}[n]e^{-j\hat{\varphi}_{P-1}(nT)} \;\; \tilde{\mathbf{d}}[n]\right]^\top. \tag{40}$$

The symbol error $e[n] = d[n] - \hat{d}[n]$ is then used to obtain the weights $\mathbf{c}[n]$ recursively over time as

$$\mathbf{c}[n+1] = \mathbf{c}[n] + \mathcal{A}\left\{\mathbf{u}[n], e[n]\right\} \tag{41}$$

where $\mathcal{A}\{\mathbf{u}[n], e[n]\}$ denotes an update term, calculated here using the recursive least squares (RLS) algorithm that operates on the input $\mathbf{u}[n]$ and the error $e[n]$. During training mode, the vector $\tilde{\mathbf{d}}[n]$ contains the training data symbols, and in the decision-directed mode, the vector $\tilde{\mathbf{d}}[n]$ contains the symbol decisions. The complete process is summarized in Algorithm 1, and a block diagram is illustrated in Fig. 2.

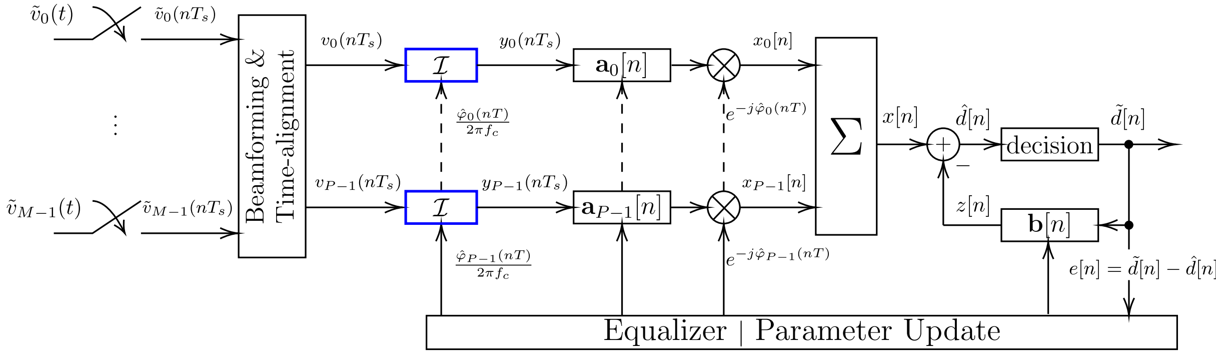


Fig. 2. Path-specific Doppler compensation receiver. Front-end beamforming is used to isolate the path contributions, which are then fed into a multichannel DFE operating in tandem with an individual PLL to track the Doppler phase of each path. The PLL output drives a DLL for each beamformed path (blue boxes). Each DLL performs delay tracking or adaptive online resampling via the linear interpolation operation $\mathcal{I}(\cdot)$ defined in (32).

V. Numerical Results

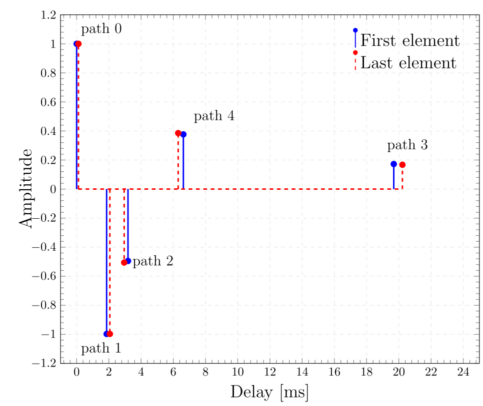
The performance of the system is assessed on a synthetic shallow water channel. The channel geometry is specified by a transmitter–receiver distance of 1 km, water depth of 100 m, receiver depth (top element) of 70 m, and transmitter depth of 20 m. The speed of sound is $c = 1500$ m/s in water, and 1300 m/s at the seabed; spherical spreading is assumed. The receiver uses a vertical array with $M = 12$ receiving elements, equally spaced by $\delta = 0.3$ m (further insights into optimal spacing are provided in [25]). The center frequency is $f_c = 12.5$ kHz, and the data rate is $R = 6500$ symbols/s. The modulation is quaternary phase-shift keying, and the noise samples are assumed to be independent identically distributed zero-mean Gaussian. The signal-to-noise ratio (SNR) is defined as the ratio between the total signal power across all receivers and the total noise power across all receivers (6).

The physical channel is illustrated in Fig. 3(a). The path amplitudes, delays, and angles are obtained from the channel geometry. In Fig. 3(b), we illustrate the power metric (18) and the relative path magnitudes $|h_p^m|$ positioned at true angles of arrival, seen on the first array element (blue solid bars) and on the last array element (red dashed bars).

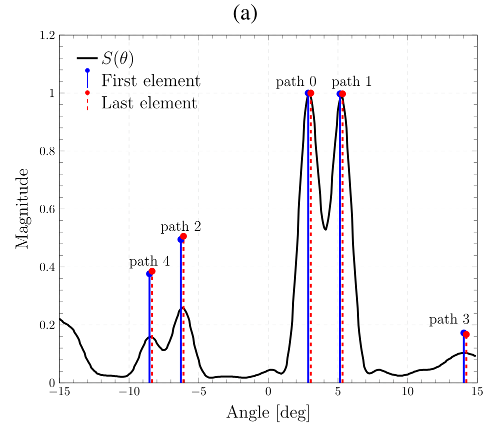
We model each propagation path using a velocity $v_{tr,p}(t)$. The corresponding Doppler factor $a_p(t)$ and delay drift $\epsilon_p(t)$ are then obtained from (27) and (26), respectively. The Doppler scenarios considered in this article are summarized in Table I. Model $\emptyset$ corresponds to the no-Doppler case, where all path velocities (and thus Doppler factors) are zero. Models I and II use constant velocities, resulting in constant Doppler factors on the order of 10^{-5} and 10^{-3}, respectively. Models III and IV employ time-varying velocities to emulate high-Doppler conditions; for these models, we list only the velocity expressions in the table for readability, noting that the corresponding Doppler factors follow directly from (27).

We consider three beamforming cases, which are summarized as follows.

1) Beamforming using two paths (BF 2-PATH): The beamformer is set to extract the two strongest paths. The two path angles are estimated and used to construct the beamforming vectors using (20).



(a)



(b)

Fig. 3. (a) Physical channel seen on the first and last array element. (b) Power metric $S(\theta)$ computed for an input SNR of 15 dB.

2) Beamforming using three paths (BF 3-PATH): The beamformer is set to extract the three strongest paths. The three path angles are estimated and used to construct the beamforming vectors using (20).

3) Beamforming using all paths (BF 5-PATH): In this case, the matrix $\hat{\mathbf{S}}_\ell'\hat{\mathbf{S}}_\ell$ becomes ill-conditioned; thus, we use the diagonally loaded null-steering matrix (22) with $\gamma = 10^{-4}$.

Algorithm 1: Multichannel Decision Feedback Equalization With Adaptive Delay Tracking.

1: **Input:** Signal samples $v_p(nT_s)$, $\forall p = 0, \ldots, P-1$, number of symbols N_d, number of feedforward and feedback equalizer filter taps N_f and N_b, number of causal filter taps N_1, number of training symbols $N_t = 4(N_f + N_b)$, training symbols $d[n]$, $n = 0, \ldots, N_t - 1$, PLL tracking coefficients K_{f_1} and K_{f_2}, forgetting factor λ, initial variance Δ^2, symbol duration T, and center frequency f_c.
2: **Initialization:** Equalizer coefficients $\mathbf{a}_p[0] = \mathbf{0}_{N_f \times 1}$, $\forall p = 0, \ldots, P-1$, and $\mathbf{b}[0] = \mathbf{0}_{N_b \times 1}$, composite equalizer filter coefficients $\mathbf{c}[0] = [\mathbf{a}_0^\top[0] \quad \ldots \quad \mathbf{a}_{P-1}^\top[0] \quad -\mathbf{b}^\top[0]]^\top$, $\mathbf{y}_p[-1] = \mathbf{0}_{N_f \times 1}, \forall p = 0, \ldots, P-1$, $\tilde{\mathbf{d}}[0] = \mathbf{0}_{N_b \times 1}$, phase estimates $\hat{\varphi}_p(0) = 0, \forall p = 0, \ldots, P-1$, $\mathbf{P}[0] = \Delta^{-2}\mathbf{I}_Q$ where $\mathbf{I}_Q$ is a $Q \times Q$ identity matrix with $Q = P \times N_f + N_b$.
3: **for** $n = 0$ to $N_d - 1$ **do**
4: **for** $p = 0$ to $P - 1$ **do**
5: $y_p(nT + i\frac{T}{2}) = \mathcal{I}\{v_p(nT + i\frac{T}{2} - \frac{\hat{\varphi}_p(nT)}{2\pi f_c})\}$, for $i = N_1, N_1 - 1$
6: $\mathbf{y}_p[n] = [y_p(nT + N_1\frac{T}{2}) \quad y_p(nT + N_1\frac{T}{2} - \frac{T}{2}) \quad \mathbf{y}_p[n-1]^\top_{1:N_f-2}]^\top$
7: $x_p[n] = \mathbf{a}'_p[n]\mathbf{y}_p[n]e^{-j\hat{\varphi}_p(nT)}$
8: **end for**
9: $\mathbf{u}[n] = [\mathbf{y}_0[n]e^{-j\hat{\varphi}_0(nT)} \quad \ldots \quad \mathbf{y}_{P-1}[n]e^{-j\hat{\varphi}_{P-1}(nT)} \quad \tilde{\mathbf{d}}[n]]^\top$
10: Feedforward output: $x[n] = \sum_{p=0}^{P-1} x_p[n]$
11: Feedback output: $z[n] = \mathbf{b}'[n]\tilde{\mathbf{d}}[n]$
12: Symbol estimate: $\hat{d}[n] = x[n] - z[n] = \mathbf{c}'[n]\mathbf{u}[n]$
13: Symbol decision:
$$\tilde{d}[n] = \begin{cases} d[n] & n < N_t \\ \text{decision}\{\hat{d}[n]\} & \text{otherwise} \end{cases}$$
14: **for** $p = 0$ to $P - 1$ **do**
15: $\psi_p[n] = \mathrm{Im}\{x_p[n][\tilde{d}[n] + z[n]]^*\}$
16: $\hat{\varphi}_p(nT + T) = \hat{\varphi}_p(nT) + K_{f_1}\psi_p[n] + K_{f_2}\sum_{i=0}^{n}\psi_p[i]$
17: **end for**
18: $e[n] = d[n] - \hat{d}[n]$
19: $\mu[n] = [\lambda + \mathbf{u}'[n]\mathbf{P}[n]\mathbf{u}[n]]^{-1}$
20: $\mathbf{c}[n+1] = \mathbf{c}[n] + \underbrace{\mu[n]\mathbf{P}[n]\mathbf{u}[n]}_{\mathbf{G}[n]} e^*[n]$
21: $\mathbf{P}[n] = \lambda^{-1}[\mathbf{I}_Q - \mathbf{G}[n]\mathbf{u}'[n]]\mathbf{P}[n-1]$
22: $\mathbf{a}_p[n+1] = \mathbf{c}[n+1]_{p \times N_f + 1:(p+1) \times N_f}$ for $p = 0, \ldots, P-1$
23: $\mathbf{b}[n+1] = -\mathbf{c}[n+1]_{Q-N_b+1:Q}$
24: $\tilde{\mathbf{d}}[n+1] = [\tilde{d}[n] \quad \tilde{\mathbf{d}}[n]_{1:N_b-1}]^\top$
25: **end for**

In Fig. 4(a) and (b), we illustrate the five path signals after beamforming, and the path signals after being time-aligned, respectively. In this case, the null-steering matrix is diagonally loaded using $\gamma = 10^{-4}$.

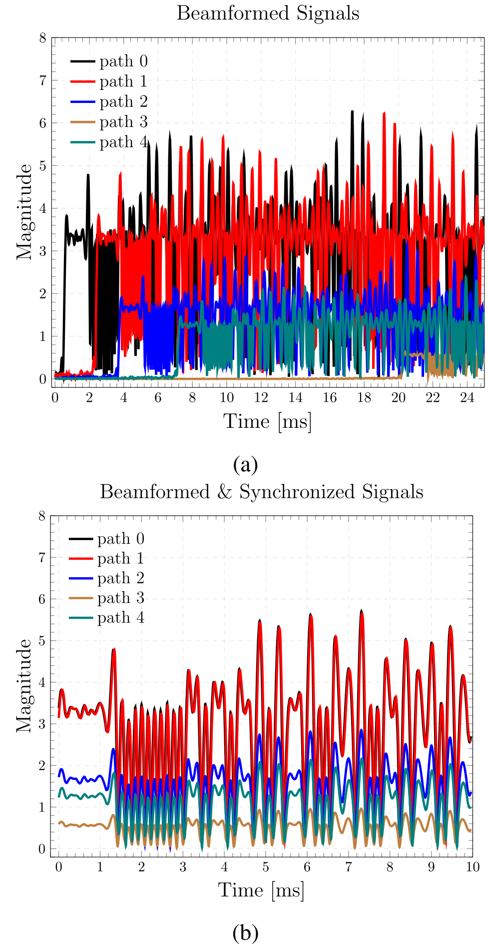


Fig. 4. Time-domain path signals (a) after beamforming and (b) after time alignment for an input SNR of 15 dB.

The time-aligned signals are subsequently fed to the DFE/PLL, which uses two fractionally spaced samples per symbol ($T/2$) and operates under the RLS algorithm with a forgetting factor of $\lambda = 0.999$ and initial variance of $\Delta^2 = 0.01$. The lengths of the feedforward and feedback equalizer filters (N_f and N_b) and the PLL coefficients (K_{f_1} and K_{f_2}) are set for each Doppler scenario (see Tables II–IV). The data detection MSE is computed as

$$\mathrm{MSE} = \frac{1}{N_r(N_d - N_t)} \sum_{i=1}^{N_r} \sum_{n=N_t}^{N_d-1} \left| d_i[n] - \hat{d}_i[n] \right|^2 \tag{42}$$

where $\hat{d}_i[n]$ is the estimate of the nth data symbol $d_i[n]$ in frame i. Each frame contains N_d data symbols, where $N_t = 4(N_f + N_b)$ symbols are used for training the DFE. An independent noise realization is used in every frame. The total number of realizations (frames) is $N_r = 20\,000$.

A. No- and Low-Doppler Regimes

We begin by evaluating system performance in the no- and low-Doppler scenarios specified by Model $\emptyset$ and Model I in Table I, respectively. A signal consisting of $N_d = 8000$ symbols is transmitted over these two channels. We compare two receiver

TABLE I
DOPPLER SCENARIOS

Model	Parameters	Path 0	Path 1	Path 2	Path 3	Path 4
$\emptyset$	velocity [m/s]	0	0	0	0	0
	Doppler factor	0	0	0	0	0
I	velocity [m/s]	0.05	-0.01	-0.05	0.025	-0.03
	Doppler factor	3.33×10^{-5}	-0.67×10^{-5}	-3.33×10^{-5}	1.67×10^{-5}	-2×10^{-5}
II	velocity [m/s]	1.5	1.6	1.4	1.55	1.65
	Doppler factor	1×10^{-3}	1.067×10^{-3}	0.93×10^{-3}	1.03×10^{-3}	1.1×10^{-3}
III	velocity [m/s]	$3 - 0.3t$	$-3 + 0.1t$	$-1 - 0.2t$	$-1.1 - 0.075t$	$1 + 0.1t$
IV	velocity [m/s]	$10 \sin(2\pi f_{v,0} t)$	$-10 \sin(2\pi f_{v,1} t)$	$-3 \sin(2\pi f_{v,2} t)$	$2 \sin(2\pi f_{v,3} t)$	$-2 \sin(2\pi f_{v,4} t)$
	$f_{v,p}$ [Hz]	0.1	0.2	0.05	0.05	0.05

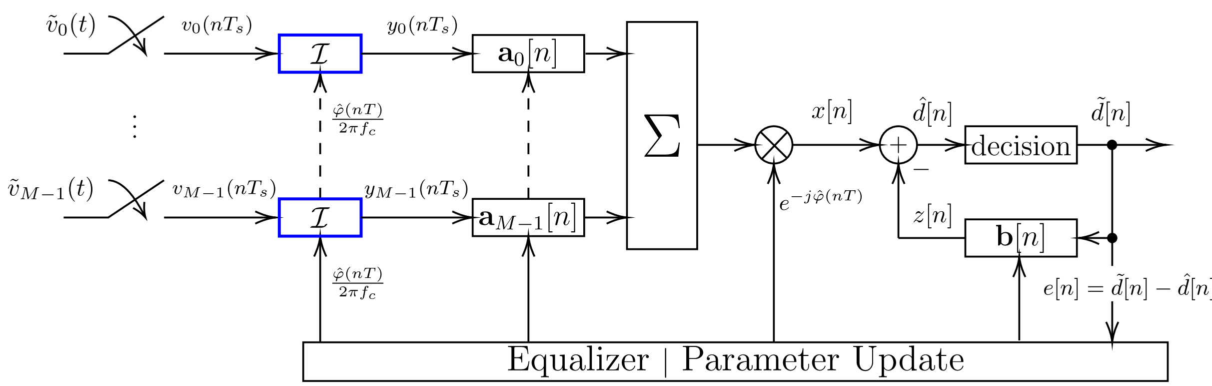


Fig. 5. Conventional receiver coupled with a single PLL to track a common Doppler phase, which drives multiple DLLs (blue boxes), one for each input signal. Each DLL performs delay tracking or adaptive online resampling via the linear interpolation operation $\mathcal{I}(\cdot)$ defined in (32).

TABLE II
DFE PARAMETERS FOR DOPPLER MODEL $\emptyset$

Algorithm	Input channels	N_f	N_b	PLL	DLL
BF p-PATH DFE	$p \in \{2, 3, 5\}$	10	10	No	No
Conventional DFE	12	10	80	No	No

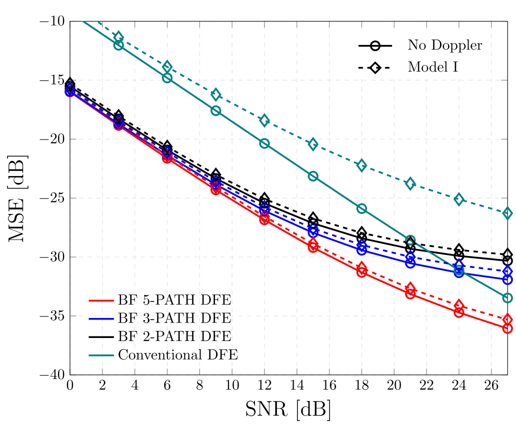


Fig. 6. Average MSE as a function of the input SNR for the path-specific Doppler and conventional methods using Model I from Table I. The path-specific Doppler techniques employ equalizer filter lengths of $N_f = 10$ and $N_b = 10$. The conventional approach uses equalizer filter lengths of $N_f = 10$ and $N_b = 80$.

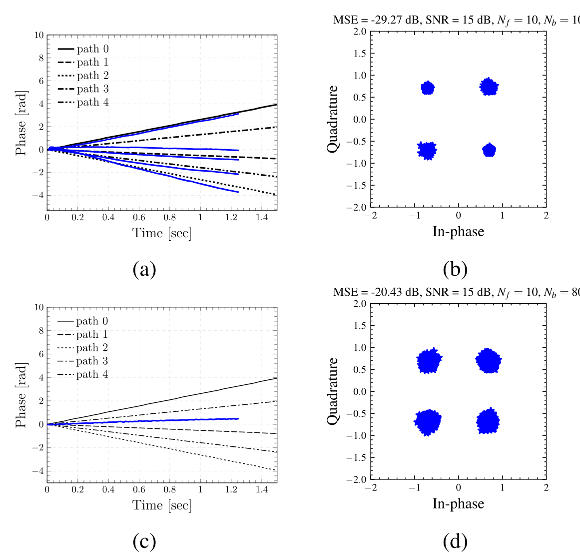


Fig. 7. (a) and (c): Estimated phases (blue lines) obtained from the PLL outputs $\hat{\varphi}_p(nT_s)$ for the BF 5-PATH DFE and Conventional DFE algorithms, respectively. The black lines show the actual phases induced on each path as specified by the Model I of Table I. (b) and (d): Scatter plot of the detected data symbols using the BF 5-PATH DFE and Conventional DFE algorithms, respectively.

systems: 1) a front-end beamforming architecture coupled with a multichannel DFE, referred to as BF p-PATH DFE, and 2) a conventional multichannel receiver without beamforming, referred to as Conventional DFE. Their block diagrams are shown in Figs. 2 and 5, respectively.

For the beamforming-based receiver, we evaluate three beamforming strategies. Each configuration is followed by a p-channel DFE, and we refer to each algorithm as BF p-PATH + DFE for $p \in 2, 3, 5$. The benchmark case, in which all 12 signals are directly fed to the equalizer, is referred to as the Conventional DFE.

TABLE III
DFE PARAMETERS FOR DOPPLER MODEL I

Algorithm	Input channels	N_f	N_b	K_{f_1}	K_{f_2}	PLL	DLL
BF p-PATH DFE	$p \in \{2, 3, 5\}$	10	10	0.01	0.001	Multiple	No
Conventional DFE	12	10	80	0.01	0.001	Single	No

TABLE IV
DFE PARAMETERS FOR DOPPLER MODELS II–IV

Model	Algorithm	Input channels	N_f	N_b	K_{f_1}	K_{f_2}	PLL	DLL
II	Conventional DFE	12	10	80	0.1	0.01	Single	No
	DLL-aided Conventional DFE	12	10	80	0.1	0.01	Single	Single
III and IV	DLL-aided BF p-PATH DFE	$p \in \{2, 3, 5\}$	6	1	0.25	0.025	Multiple	Multiple

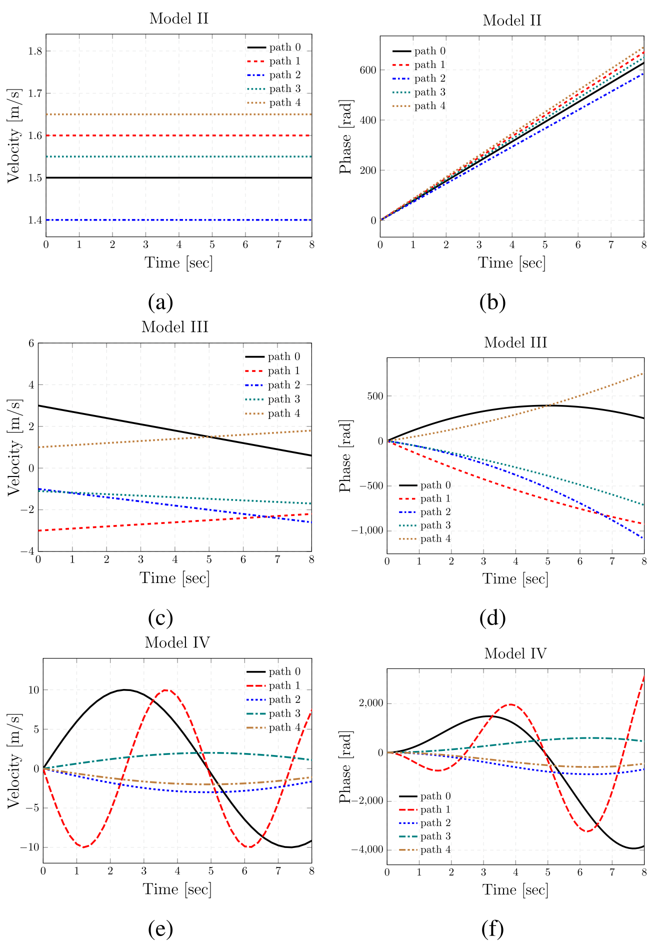


Fig. 8. (a)–(c): Relative velocities experienced on each path, $v_{tr,p}(t)$, for high-Doppler regimes (Models II–IV of Table I). (d)–(f): Doppler-induced phases, $2\pi f_c \epsilon_p(t)$, where $\epsilon_p(t) = a_p(t) \cdot t$ is the Doppler drift and $a_p(t) = (v_{tr,p}(t))/c$ is the Doppler factor experienced on each path.

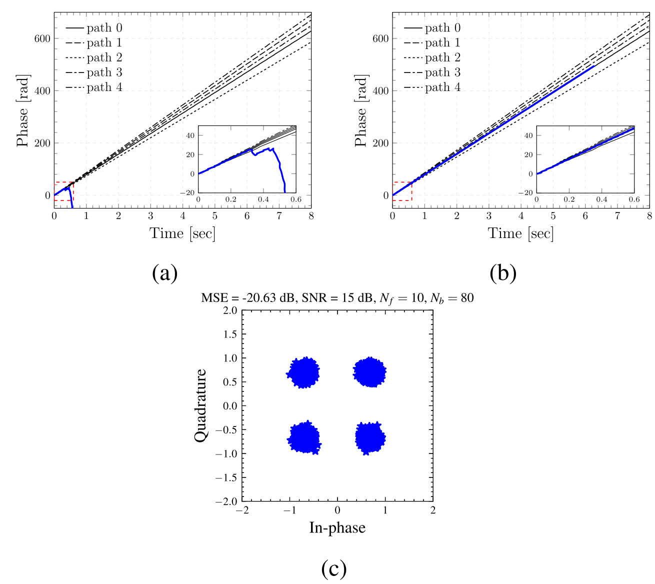


Fig. 9. Estimated phases (blue lines) obtained from the PLL outputs $\hat{\varphi}(nT_s)$ for the following receiver configurations: (a) Conventional DFE and (b) DLL-aided Conventional DFE. The black curves show the actual phases induced on each path as specified by the Model II of Table I. (c) Scatter plot of the detected data symbols using the DLL-aided Conventional DFE.

In the no-Doppler scenario, neither receiver employs a PLL nor a DLL, and the DFE parameters are specified in Table II. In the low-Doppler scenario, the DFE parameters for both receivers are given in Table III. The beamforming methods use multiple PLLs to track the Doppler shifts on individual multipath components, whereas the conventional approach utilizes a single PLL. No DLL is employed in either case.

The system performance is summarized in Fig. 6 in terms of the data detection MSE, for an input SNR that ranges from 0 to 27 dB. The front-end beamforming methods utilize a narrow DFE field of view ($N_f = 10$ and $N_b = 10$), while the conventional approach uses equalizer filters lengths of $N_f = 10$ and $N_b = 80$.

For the no-Doppler scenario (curves marked with ○), the lowest MSE is achieved by the BF 5-PATH DFE technique across all SNR values. When insufficient spatial resolution prevents beam placing toward all the paths, BF 2-PATH DFE and BF 3-PATH DFE techniques exhibit an MSE performance degradation of 1–6 dB for input SNRs between 12 and 27 dB. In both cases, the MSE saturates at approximately −29 dB for SNRs above 21 dB. This saturation occurs because the contribution of the remaining paths is neglected.

The highest MSE corresponds to the Conventional DFE method, which shows a degradation in MSE ranging from 7 to 5 dB with respect to the BF 2-PATH DFE and BF 3-PATH DFE cases for input SNRs up to 15 dB. For SNRs above 20 dB,

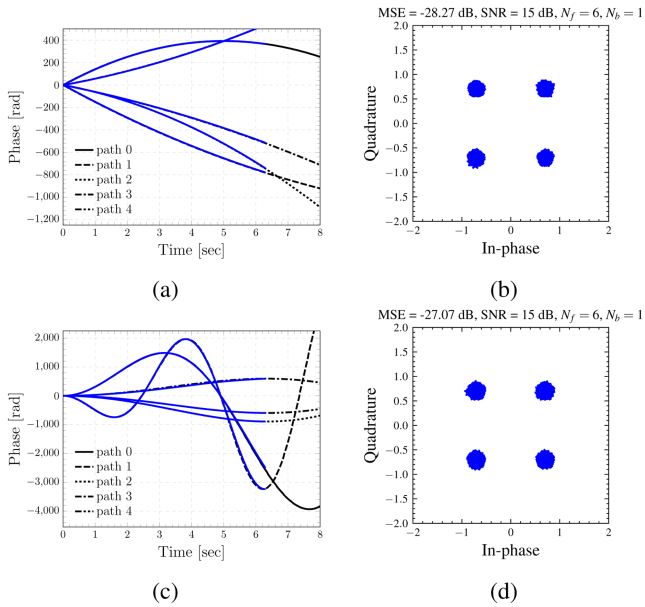


Fig. 10. Estimated phases (blue lines) and the corresponding scatter plot of estimated data symbols for Doppler Models III (a) and (b) and IV (c) and (d). The curves shown in black correspond to the true path phases. The receiver configuration is the DLL-aided BF 5-PATH DFE for an input SNR of 15 dB.

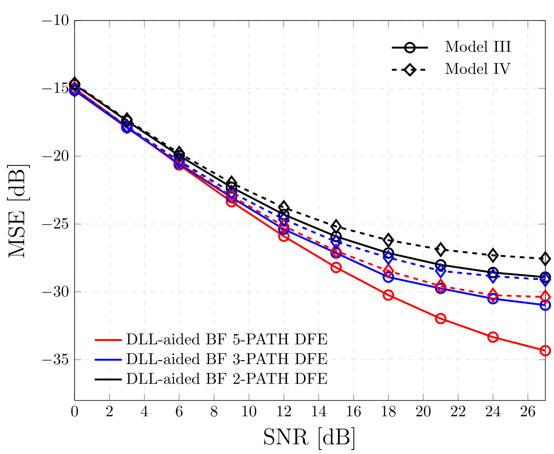


Fig. 11. Average MSE as a function of the input SNR for the path-specific Doppler method using Models III and IV from Table I. The conventional receiver fails under these high-Doppler regimes.

the Conventional DFE outperforms the BF 2-PATH DFE and BF 3-PATH DFE techniques, both of which exhibit saturation. Nevertheless, the Conventional DFE still shows a degradation in MSE ranging from 8 to 3 dB compared with the BF 5-PATH DFE technique.

For the low-Doppler regime (curves marked with $\diamond$), the front-end beamforming methods, BF p-PATH DFE for $p \in \{2, 3, 5\}$, suffer a degradation in MSE of approximately 0.5–1 dB compared with their performance in the no-Doppler case, across all SNR values. In contrast, the Conventional DFE method (green dotted lines) exhibits a degradation of 1–6 dB in MSE relative to its performance in the no-Doppler regime, for the full range of SNR values.

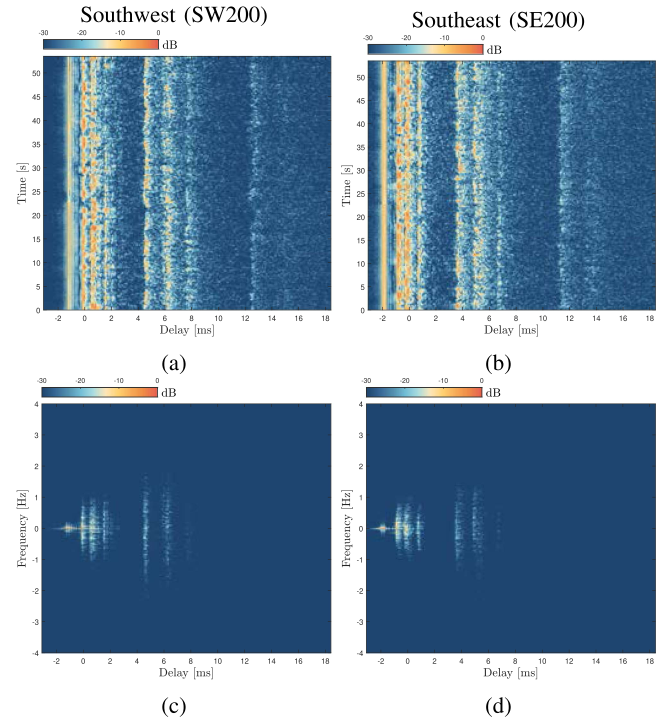


Fig. 12. SPACE: (a) and (b) time-delay channel impulse responses and (c) and (d) the corresponding delay–Doppler responses observed at the top element of the array.

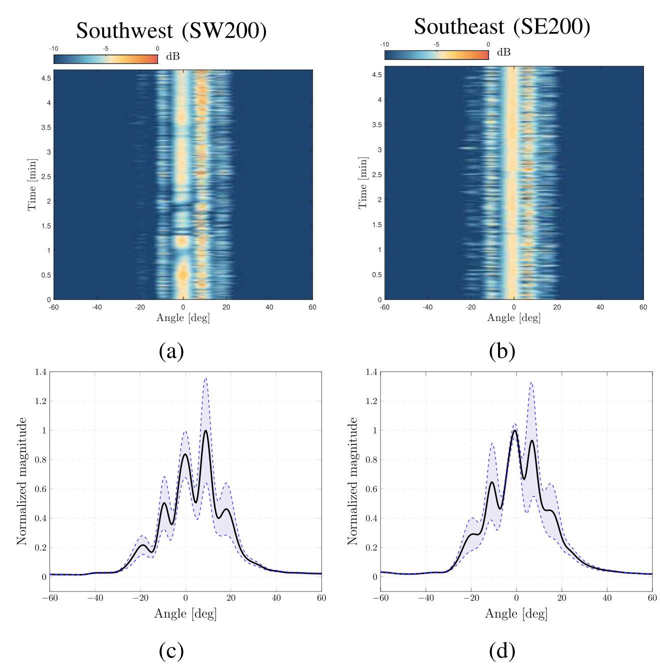


Fig. 13. SPACE: (a) and (b) Power metrics (18) across time in dB with respect to the peak value. (c) and (d) Power metrics averaged across time (black lines).

This result demonstrates that the proposed algorithm is capable of tracking the path-specific Doppler drifts while using only a few equalizer filter taps, as opposed to the conventional approach, which necessitates a considerably longer equalizer.

In addition, the estimated phases obtained from the PLL outputs and constellations of the detected data symbols are

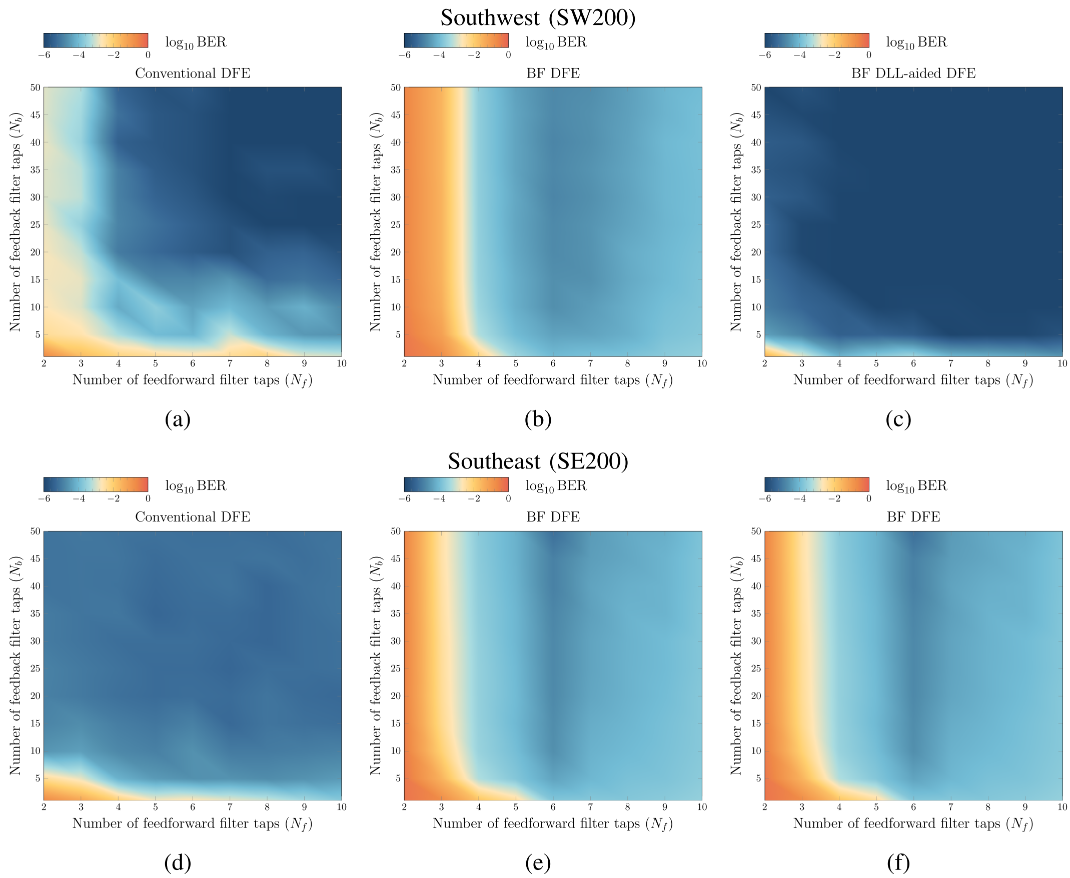


Fig. 14. SPACE: Average BER as a function of the number of feedforward and feedback taps N_f and N_b for the Conventional DFE method (a) and (d), the BF DFE (b) and (e), and the BF DLL-aided DFE (c) and (f).

illustrated in Fig. 7. The top row corresponds to the BF 5-PATH DFE configuration, while the bottom row depicts the results corresponding to the Conventional DFE. The estimated Doppler phases from the PLL outputs of the BF 5-PATH DFE technique, depicted in Fig. 7(a), show that front-end beamforming can effectively isolate and track the path-specific Doppler drifts with only a few equalizer taps. On the contrary, the estimated phase of the Conventional DFE approach, illustrated in Fig. 7(c), can only track a common part of the Doppler drifts, leaving the residual per-path Doppler drifts to the equalizer. In Fig. 7(b) and (d), we illustrate the constellations of the estimated data symbols, where the BF 5-PATH technique delivers an MSE of approximately -29 dB compared to -20 dB that is obtained by using the Conventional DFE technique for an input SNR of 15 dB.

B. High-Doppler Regimes

In this section, we analyze the high-Doppler regimes specified by Models II–IV. A signal that contains $N_d = 40\,000$ symbols and spans a duration of approximately 6 s is transmitted using these three channel models, where we consider the relative velocities $v_{tr,p}(t)$ experienced on each path, and the corresponding Doppler-induced phases, $2\pi f_c \epsilon_p(t)$, are shown in Fig. 8.

It is observed that the Conventional DFE receiver fails in all regimes due to the high path-specific Doppler variation and long signal duration. The inclusion of an adaptive resampling mechanism allows the conventional receiver, referred to as DLL-aided Conventional DFE, to track a common Doppler-induced phase of Model II. However, the DLL-aided Conventional DFE fails under the Models III and IV. In contrast, front-end beamforming with path-specific Doppler tracking, referred to as DLL-aided BF p-PATH DFE, operates successfully in all regimes. Detailed results, obtained from the system configurations listed in Table IV, are shown below.

In Fig. 9, we illustrate the performance of the conventional receiver with and without adaptive delay tracking, referred to as Conventional DFE and DLL-aided Conventional DFE, respectively. Both receiver configurations use an RLS algorithm with $\lambda = 0.999$ and $\Delta^2 = 0.01$. The Conventional DFE receiver fails to properly track the Doppler-induced phase after 0.5 s, as seen in Fig. 9(a). The failure is caused by the signal slipping outside of the equalizer's window. However, the DLL-aided Conventional DFE receiver tracks the common Doppler-induced phase for the entire duration of the signal while keeping the same equalizer filter lengths. A data detection MSE of -20 dB for an input SNR of 15 dB is delivered. The estimated phase (blue line) obtained from the PLL output and the scatter plot of the detected data symbols are illustrated in Fig. 9(b) and (c), respectively.

The performance of DLL-aided BF p-PATH DFE algorithms for Models III and IV is illustrated in Fig. 10. In contrast to the conventional receiver configurations, which fail in these cases,

TABLE V
SPACE, MACE, AND KAM SIGNAL PARAMETERS

	SPACE	MACE	KAM
center freq. f_c [kHz]	12.5	13	13
sampling freq. f_s [kHz]	$10^7/256$	$10^7/256$	$10^7/256$
symbol rate R [symbols/s]	6510.4	4882.8	6250
M-sequence length	$2^{12}-1$	$2^{11}-1$	$2^{12}-1$
modulation	BPSK	BPSK	BPSK
Julian day	288	176 and 177	188
bits processed	1760850	47314358	331776

we observe a superior MSE performance with approximately −28 and −27 dB for an input SNR of 15 dB (observe the scatter plots of the detected data symbols in Fig. 10(b) and (d). In addition, the estimated phases (blue lines) obtained from the PLL outputs are illustrated in Fig. 10(a) and (c), which show that the front-end beamforming with adaptive online resampling algorithms is able to successfully track the path-specific Doppler-induced phases.

The data detection MSE performance of the configurations listed in Table IV is summarized in Fig. 11. The general trend is analogous to the one observed before, indicating that front-end beamforming with per-path adaptive online resampling is able to successfully track the time-varying path-specific Doppler drifts of Fig. 8(d) and (f). We emphasize that the conventional receiver fails under the Models III and IV.

VI. Experimental Results

In this section, we provide the results of processing real data recorded during three at-sea experiments: the SPACE, the MACE, and the MURI KAM, conducted in 2008, 2010, and 2011, respectively. These experiments encompass different channel geometries, receiver array geometries, and signal parameters (specified in Table V).

A. *Surface Processes Acoustic Communications Experiment*

The SPACE experiment took place approximately 3 km south of Martha's Vineyard, MA, USA, in October 2008 [26]. The data presented here were recorded on an array located at approximately 200 m southwest (SW200) and 200 m southeast (SE200) of the source. The receiver array consisted of 24 elements equally spaced by 5 cm and was deployed 3.3 m above the seafloor. The transmitter was deployed at 4 m above the seabed, and the water depth was 15 m. During the experiment, each transmission lasted 1 min and included 86 repeated binary phase-shift keying (BPSK) modulated M-sequences of length 4095, whose parameters are presented in Table V. The time-delay impulse responses and delay–Doppler responses of the channel, estimated from a transmission during Julian day 288 [27], are illustrated in Fig. 12 for the SW200 and SE200 receiver systems.

The proposed algorithm is applied to five consecutive experimental signal transmissions that were simultaneously recorded on systems SW200 and SE200, during the Julian day 288. Each transmission is processed in blocks of 4095 symbols. We assess the performance of three systems: A conventional 24-channel DFE with a single PLL, a beamforming-based system with per-path Doppler tracking with and without adaptive resampling. For the beamforming systems, the power metric (18) used to estimate the angles of arrival on each block is plotted across time in Fig. 13(a) and (b) for the receiver systems SW200 and SE200, respectively. There are five path angles identified by the array SW200 and three path angles by the array SE200 [see Fig. 13(c)

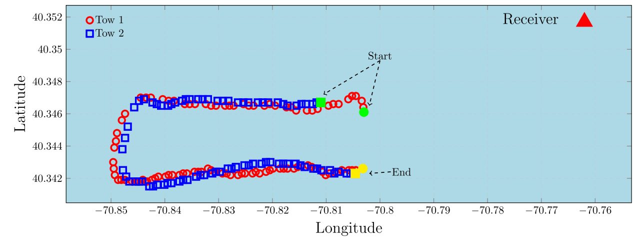


Fig. 15. Ship trajectory during the MACE experiment. There were 99 transmissions during Tow 1 (black circles) and 83 transmissions during Tow 2 (blue squares). The green- and yellow-colored points show the start and the end of the transmitter trajectories, respectively. The transmitter moved away from the receiver and then toward the receiver at a varying speed that ranged from 0.5 to 1.5 m/s. Each transmission spanned a total time of 1 min and consisted of repeated binary PSK (BPSK) M-sequences of length 2047.

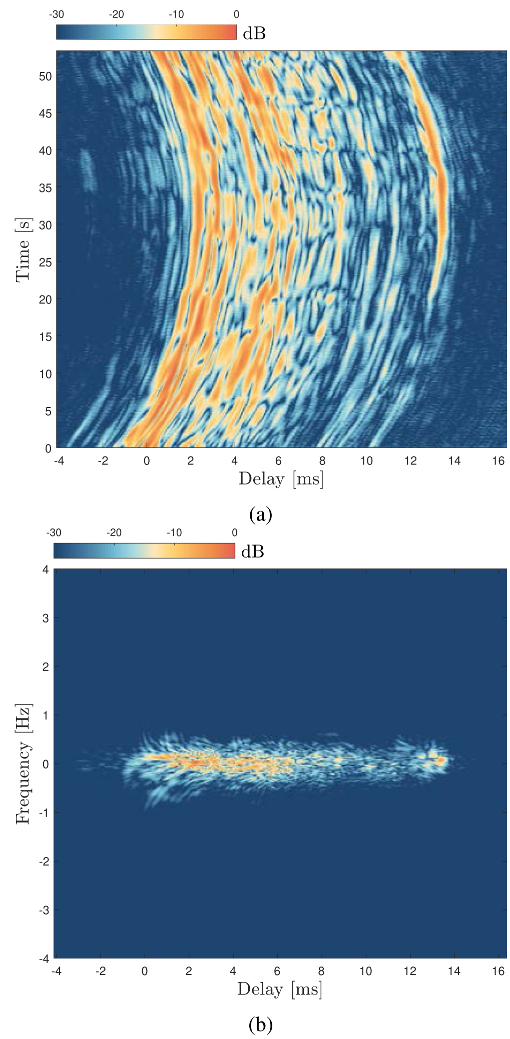


Fig. 16. MACE: (a) time-delay channel impulse response and (b) corresponding delay–Doppler response observed at the top element of the array.

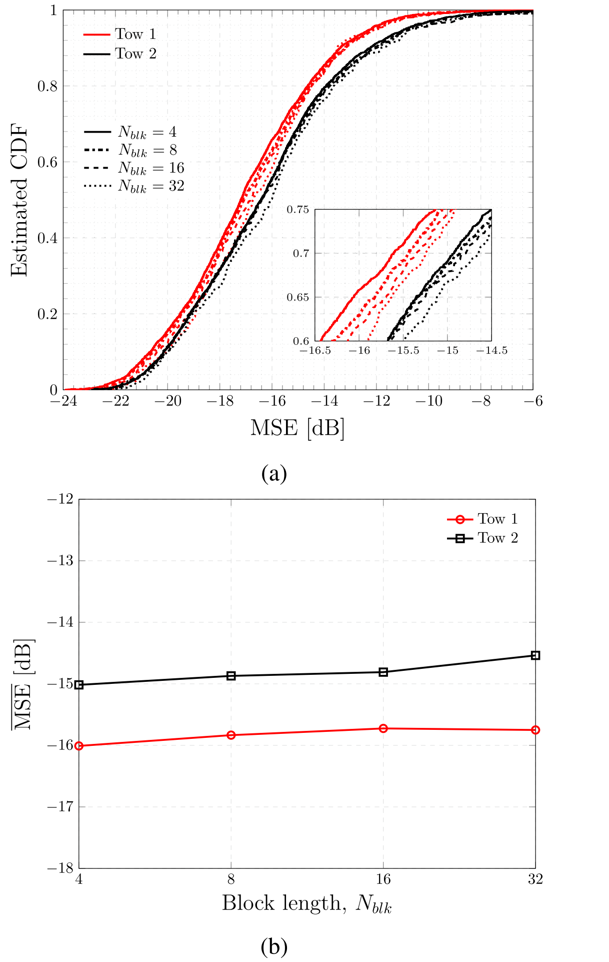


Fig. 17. MACE: (a) Estimated CDF of the MSE obtained by applying the proposed algorithm to the signals of tow 1 (red lines) and tow 2 (black lines), using various block lengths, $N_{\mathrm{blk}} = 4, 8, 16, 32$. The equalizer filter lengths are $N_f = 10$ and $N_b = 50$. (b) Average MSE as a function of the block length N_{blk} for tows 1 and 2.

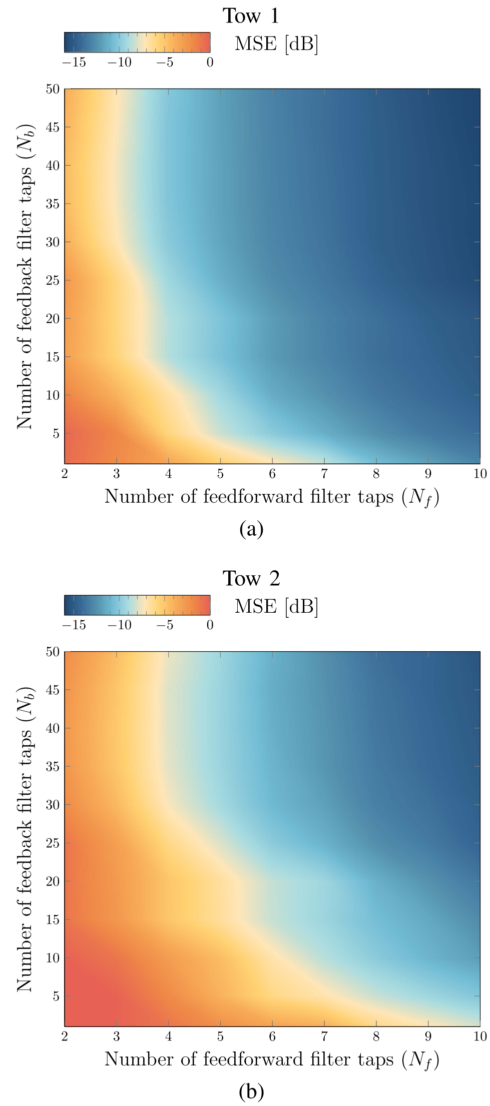


Fig. 18. MACE: Average MSE as a function of N_f and N_b for tows 1 (left) and 2 (right).

and (d), respectively]. The beamformed paths are subsequently fed to a multichannel equalizer. The DFE is equipped with an RLS algorithm with forgetting factor of $\lambda = 0.999$, initial variance of $\Delta^2 = 0.01$, and the PLL tracking constants are set to $K_{f_1} = 10^{-4}$ and $K_{f_2} = 10^{-5}$ in all cases.

The average BER as function of various filter length pairs (N_f, N_b) is the metric employed to evaluate the system performance. The performance delivered by the SW200 receiver is illustrated in the top panel of Fig. 14, where the conventional method delivers average BERs of as low as 1×10^{-6} for the region defined approximately between $7 \leq N_f \leq 10$ and $25 \leq N_b \leq 50$. The front-end beamforming method delivers BERs of only 1×10^{-5} by using filters of lengths $N_f = 6$ and $15 \leq N_b \leq 50$. However, the front-end beamforming with adaptive resampling achieves BERs of 1×10^{-6} for the region approximately limited between $4 \leq N_f \leq 10$ and $10 \leq N_b \leq 50$. These results show that the front-end beamforming system equipped with adaptive path-specific Doppler tracking delivers a performance similar to that of the conventional receiver, but with fewer equalizer filter taps. The front-end beamforming receiver requires the DFE to update a total of $5 \times 4 + 10 = 30$ filter coefficients, while the conventional system requires $24 \times 7 + 25 = 193$ filter taps. Similar performance trends are delivered by the system SE200 and illustrated in the bottom panel of Fig. 14.

B. Mobile Acoustic Communication Experiment

The MACE experiment was carried out off the coast of Martha's Vineyard, MA, USA, in June 2010. The receiver array, consisting of 12 equally spaced elements and spanning a total linear aperture of 1.32 m, was deployed at a depth of 40 m, and the transmitter moved away and toward the receiver at varying speeds of approximately 0.5–1.5 m/s, as illustrated in Fig. 15. The transmitter station was towed at the depth of 40–60 m during Julian days 176 (tow 1) and 177 (tow 2). The water depth was approximately 100 m, while the transmission distance varied between 3 and 7 km. In this experiment, each transmission spanned approximately 1 min and consisted of 128

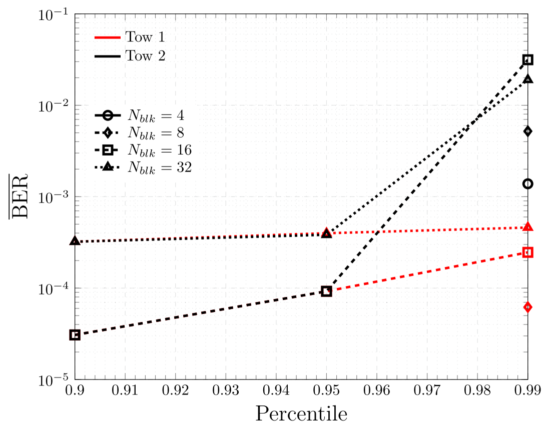
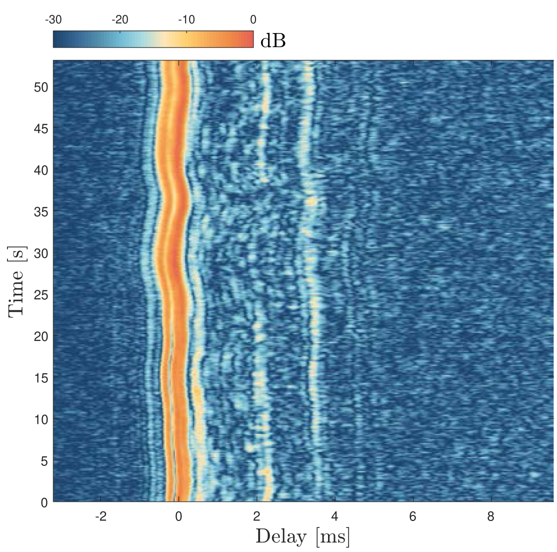
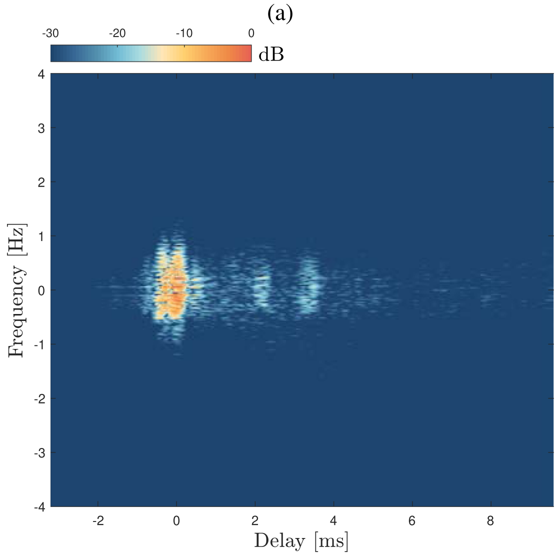


Fig. 19. Average BER for various percentiles.

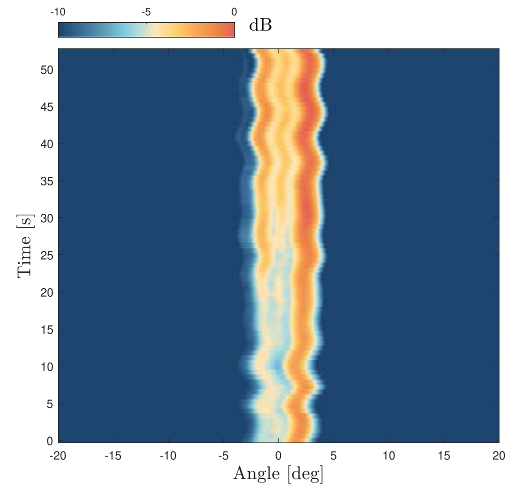


(a)

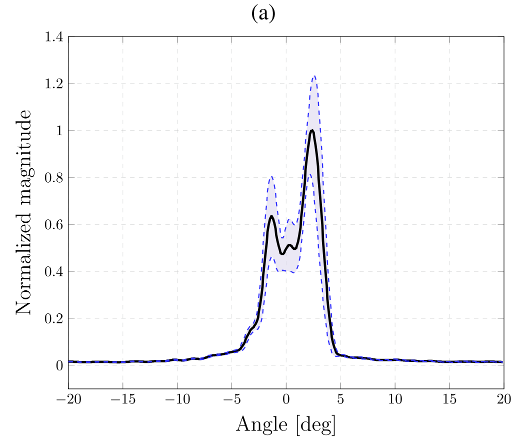


(b)

Fig. 20. KAM: (a) time-delay channel impulse response and (b) the corresponding delay–Doppler response observed at the top element of the array.



(a)



(b)

Fig. 21. KAM: (a) Power metric (18) across time. (b) Average power metric.

repeated BPSK modulated M-sequences of length 2047 whose parameters are presented in Table V. There were a total of 99 and 81 transmissions during tows 1 and 2, respectively. The time-delay channel impulse response and the delay–Doppler response, estimated from a transmission during tow 2 [27], are illustrated in Fig. 16.

In this case, we evaluate the performance of a conventional receiver equipped with adaptive online resampling based on linear resampling. In other words, the 12 incoming signals are adaptively resampled and then fed to a multichannel DFE with a single PLL. To assess the performance of the proposed algorithm, we process blocks of M-sequences of length $N_d = 2047$. Each superblock contains $N_d^b = N_d \times N_{\text{blk}}$ data symbols. We demonstrate the system performance in terms of data detection MSE for varied number of blocks $N_{\text{blk}} = 4, 8, 16, 32$, spanning a total time of 1.677, 3.354, 6.708, 13.415 s, respectively. The data detection MSE of the ith superblock is calculated as

$$\mathrm{MSE}_i^b = \frac{1}{N_d^b - N_t} \sum_{n=N_t}^{N_d^b - 1} \left| d_i[n] - \hat{d}_i[n] \right|^2 \tag{43}$$

and, since the channel varies randomly, MSE_i^b is a random variable with average

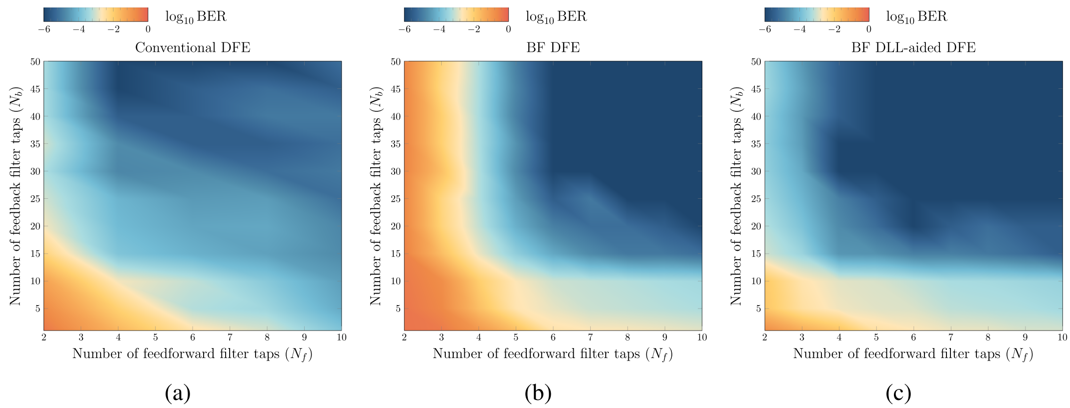


Fig. 22. KAM: Average BER as a function of N_f and N_b for (a) the Conventional DFE, (b) the BF DFE, and (c) the BF DLL-aided DFE.

$$\overline{\mathrm{MSE}}^b = \frac{1}{N_a}\sum_{i=1}^{N_a}\mathrm{MSE}_i^b \tag{44}$$

where N_a is the total number of superblocks.

The DFE is equipped with an RLS algorithm with a forgetting factor of $\lambda = 0.999$, initial variance of $\Delta^2 = 0.01$, and the PLL coefficients are $K_{f_1} = 0.025$ and $K_{f_2} = 0.0025$. Fig. 17(a) illustrates the estimated cumulative distribution function (CDF) of the MSE (43) for tow 1 (continuous lines) and tow 2 (dotted lines) for various superblock lengths. For 70% of the blocks of lengths $N_{\text{blk}} = 4, 8, 16, 32$, the proposed algorithm delivers an MSE below -15.6, -15.43, -15.35, and -15.25, respectively, during tow 1. Similarly, an MSE below -14.92, -14.85, -14.79, and -14.64 dB is obtained during tow 2, respectively.

Fig. 17(b) depicts the average MSE for various block lengths during tows 1 and 2. A degradation of approximately $0.4-0.5$ dB in the MSE is observed between minimum and maximum lengths of the superblocks, for both tows 1 and 2. Although performance degradation is expected due to the loss of the temporal coherence of the channel, this small degradation of MSE strongly supports in favor of the proposed algorithm, demonstrating its capabilities to track the Doppler over extended periods. Note that the filter lengths are kept the same regardless of the total time duration.

Fig. 18 shows the average MSE as a function of the filter lengths N_f and N_b for $N_{\text{blk}} = 4$. The system delivers excellent average data detection MSE of approximately -16 and -15 dB for tow 1 [see Fig. 17(a)] and tow 2 [see Fig. 17(b)], and using a DFE window determined by $N_f = 10$ and $N_b = 50$.

Similarly, the average BERs for different percentiles are plotted in Fig. 19. These results are obtained by processing a total of 25 736 931 and 21 577 427 bits during tows 1 and 2, respectively. For 90% of the blocks of length $N_d \times N_{\text{blk}}$ with $N_{\text{blk}} = 16$ and 32, the system delivers an average BER of approximately 3×10^{-5} and 3×10^{-4}, respectively, during tows 1 and 2. For block lengths of $N_{\text{blk}} = 4$ and 8, no bit errors are observed. For 95% of the blocks with $N_{\text{blk}} = 16$ and 32, the system reports an average BER of 9.2×10^{-5} and 3.8×10^{-4}, respectively, during tows 1 and 2. No bit errors are delivered for block lengths of $N_{\text{blk}} = 4$ and 8.

C. KAM

The KAM experiment was conducted in July 2011 off the coast of Kauai, HI, USA. Both the transmitter and the receiver were deployed approximately in the middle of a water column of 100 m and were 7 km apart. The receiver included an array of 24 elements equally spaced by 0.2 m (spanning a total linear aperture of 4.8 m). During the experiment, each transmission consisted of 81 repeated BPSK modulated M-sequences of length 4095.

The time-delay channel impulse response and the delay–Doppler response, estimated from a transmission during Julian day 188 [27], are illustrated in Fig. 20.

A single experimental transmission that was recorded during the Julian day 188 is processed in blocks of 4095 symbols. We analyze the performance of three systems: A conventional 24-channel DFE with a single PLL, a beamforming-based system with per-path Doppler tracking with and without adaptive resampling. For the beamforming systems, the power metric (18) used to estimate the angles of arrival on each block is plotted across time in Fig. 21(a), where two path angles are identified [see Fig. 21(b)]. The two beamformed paths are subsequently fed to a multichannel equalizer. The DFE is equipped with an RLS algorithm with forgetting factor of $\lambda = 0.999$, initial variance of $\Delta^2 = 0.01$, and the PLL tracking constants are set to $K_{f_1} = 10^{-4}$ and $K_{f_2} = 10^{-5}$ in all cases.

The average BER performance is computed for various filter length pairs (N_f, N_b) and shown in Fig. 22. The conventional method delivers average BERs of as low as 1×10^{-5} for the region defined approximately between $4 \leq N_f \leq 10$ and $40 \leq N_b \leq 50$. The front-end beamforming method delivers BERs of 1×10^{-5} by using filters of lengths $5 \leq N_f \leq 10$ and $15 \leq N_b \leq 50$. The front-end beamforming with adaptive resampling achieves BERs of 1×10^{-5} for the region limited between $4 \leq N_f \leq 10$ and $15 \leq N_b \leq 50$. In this case, the beamforming approaches requires the DFE to update a total of $2 \times 4 + 15 = 23$ filter coefficients, while the conventional system requires $24 \times 4 + 40 = 136$ filter taps.

VII. Conclusion

In this article, we presented a high-rate information transmission scheme for underwater channels. The framework includes a receiver equipped with a uniform linear array. Beamforming is used to isolate multiple propagation paths, allowing each path to be processed independently, and eliminates the need for long equalizers. Ideally, each beamformed path requires a single-tap equalizer, or practically, a few fractionally spaced taps. The multiple path components are time aligned and then recombined using a multichannel DFE. In this manner, the DFE is capable of tracking the Doppler-induced phase for each path individually. In addition, we introduced a DLL in tandem with the PLL to perform adaptive online resampling. The DLL can be implemented in the context of a conventional multichannel receiver and the proposed front-end beamforming receiver. The proposed algorithm operates successfully in Doppler scenarios, where the conventional receiver fails. Moreover, in cases where the conventional receiver performs well, the proposed algorithm achieves comparable performance with significantly fewer training symbols. We demonstrated the system performance using synthetic data and experimental transmissions obtained from the SPACE, MACE, and KAM data sets. Judging by our results in terms of data detection MSE and BER, the front-end beamforming system provides successful solution to the long-standing path-specific Doppler problem.

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